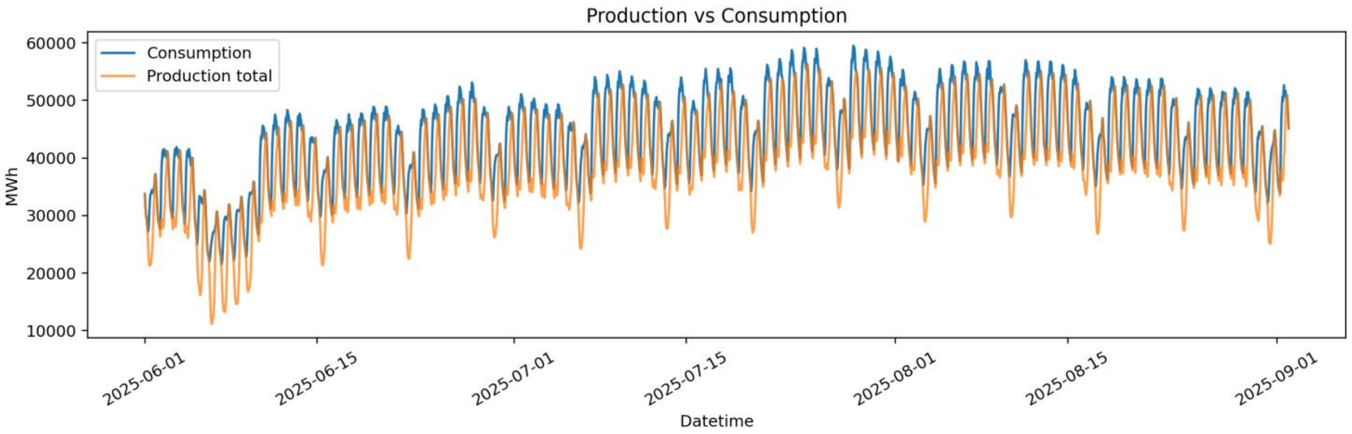
Electricity Consumption Regression

Decision Tree and XGBoost models for hourly load prediction

MEE344 Machine Learning – Term Project

Group Members:
Yusuf Ali Demirkıran **200029900**
Fatih Umut Sadıkoğlu **200030322**
Batuhan Çanga **Y23LELE001**
Doğukan Akhan **200027062**
Bahadır Çetin **200026717**

Dataset window: 1 Jun 2025 - 1 Sep 2025
Target: hourly electricity consumption (Tuketim)



Problem Definition

A supervised regression task with explicit forecasting constraints

We predict hourly electricity consumption. The target is continuous, so the task is regression.

Same-hour production is excluded from the main model because it would not be known before forecasting that hour.

The model uses past consumption, past production and known calendar information.

Dataset Overview

Two hourly CSV files are merged by timestamp

2232
raw production rows

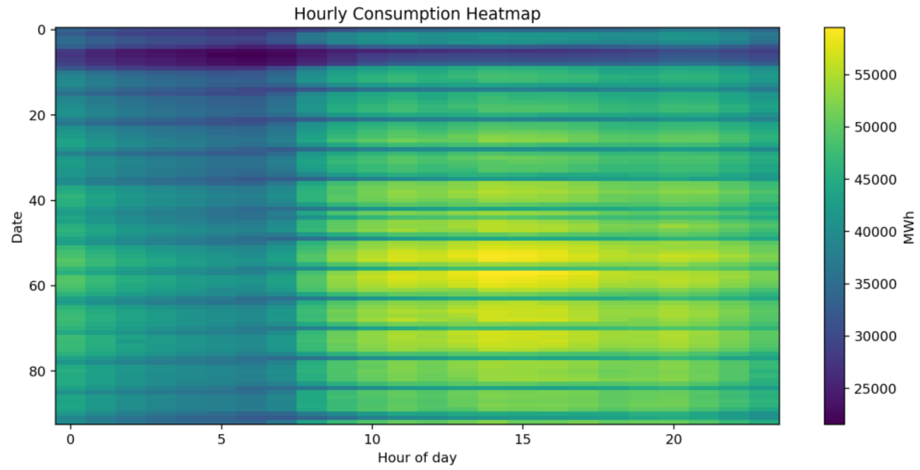
2232
raw consumption rows

2064
modeled samples

309
test samples

37
features used

Item	Status
Time range	2025-06-01 00:00 to 2025-09-01 23:00
Frequency	Hourly, no gaps detected
Merge key	datetime
Target	Tuketim (MWh)
Constant columns removed	Nafta, LNG



Preprocessing Decisions

Data cleaning was kept explicit and reproducible

Parse Turkish number formats
33.758,78 -> 33758.78

→

Create datetime
from Tarih + Saat

→

Merge production + consumption
by timestamp

→

Remove constant columns
Nafta, LNG

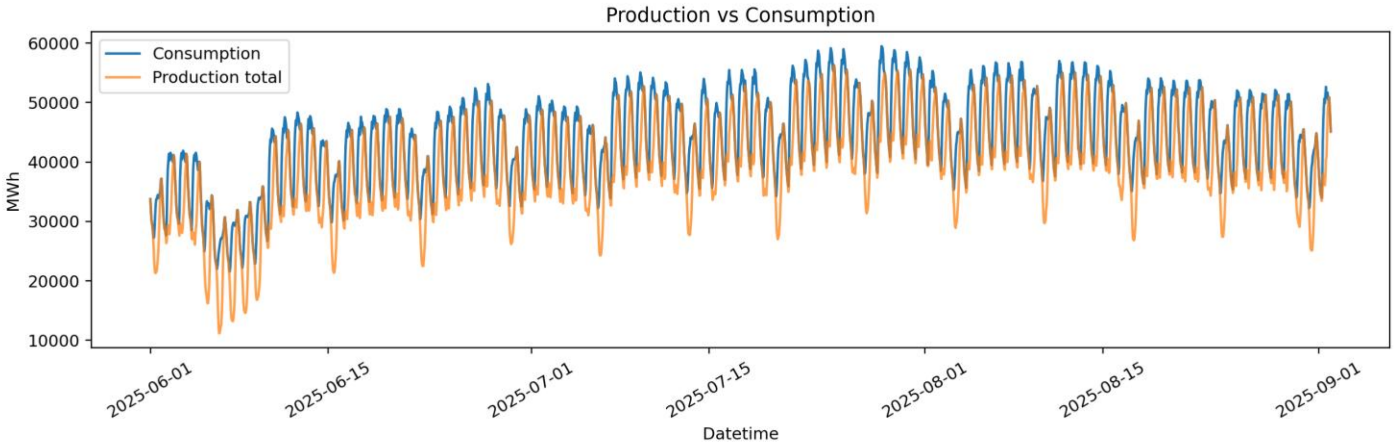
No raw missing values were found. Missing rows introduced by lag/rolling features are dropped after feature construction.

No nominal categorical variables were present. Date/time strings are encoded as numeric calendar and cyclic features.

No standardization is applied: Decision Tree and XGBoost are tree-based methods and do not require feature scaling.

Exploratory Analysis: System Load

Consumption and production are correlated but not identical

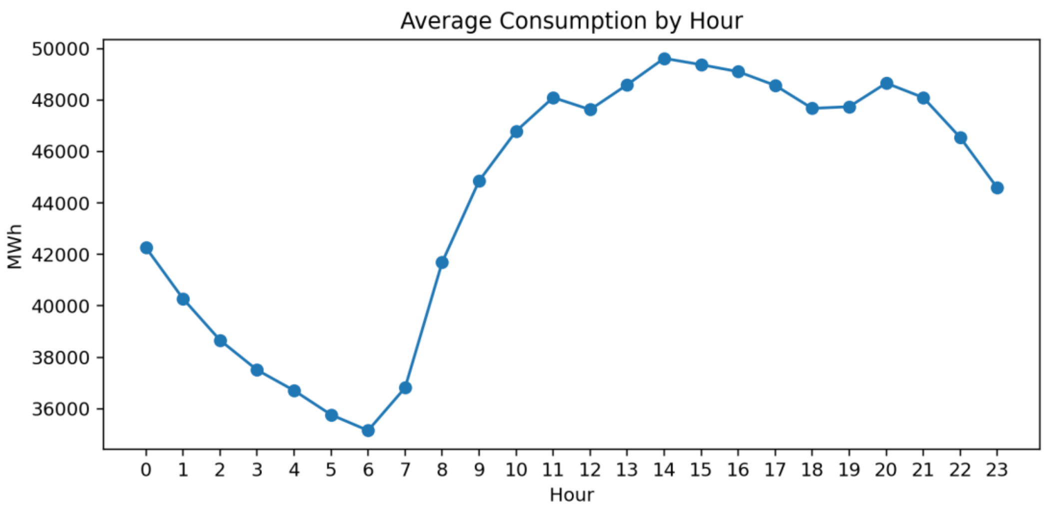


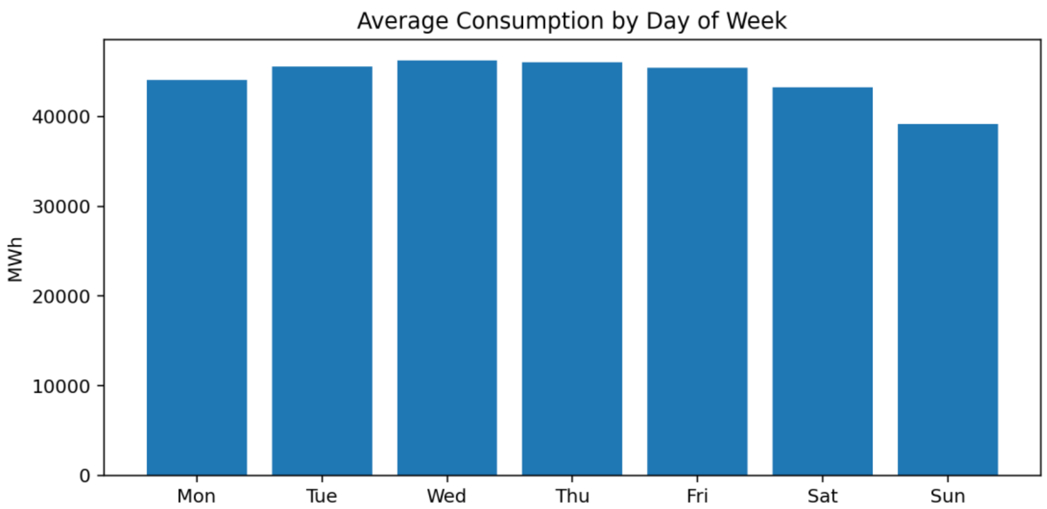
Observation	Implication
Production total is not equal to consumption	Do not treat Toplam as target substitute
Consumption follows daily demand cycles	Use calendar features
Lag autocorrelation is strong	Use lag and rolling features

Key modeling choice: same-time production values are excluded from the main forecast to avoid feature availability leakage.

Exploratory Analysis: Temporal Patterns

Hourly and weekly structure justify calendar features



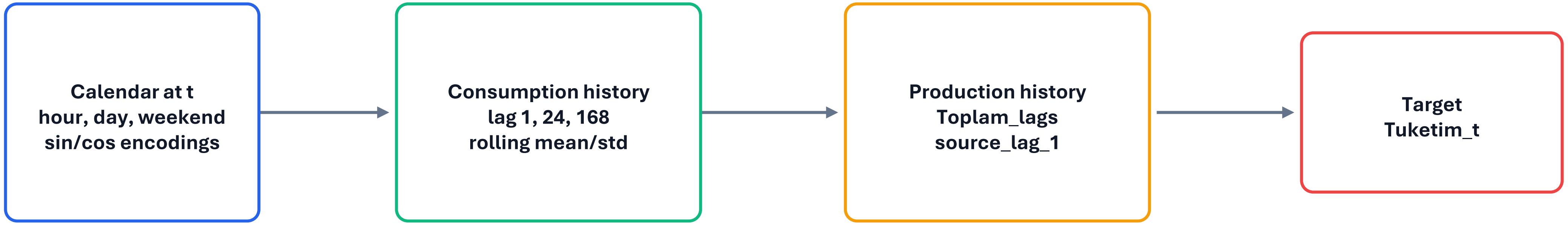


Average consumption rises after early morning and remains high during daytime/evening.

Weekends, especially Sunday, show lower average consumption than weekdays.

Feature Engineering

Rows represent prediction time t; features use t-1 or earlier

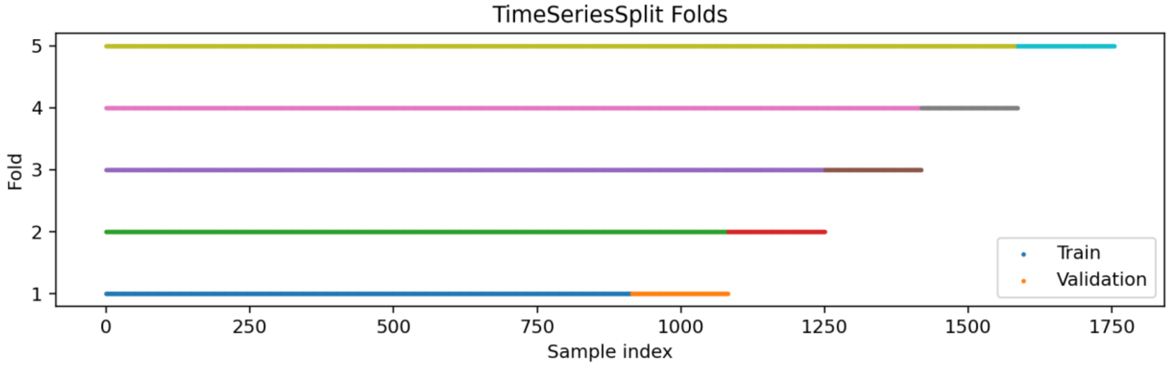


Lag 1: previous hour. Lag 24: same hour yesterday. Lag 168: same hour last week.

Rolling features are computed as shift(1).rolling(...), so the target row is never included.

Validation Methodology

The test period is untouched until final evaluation



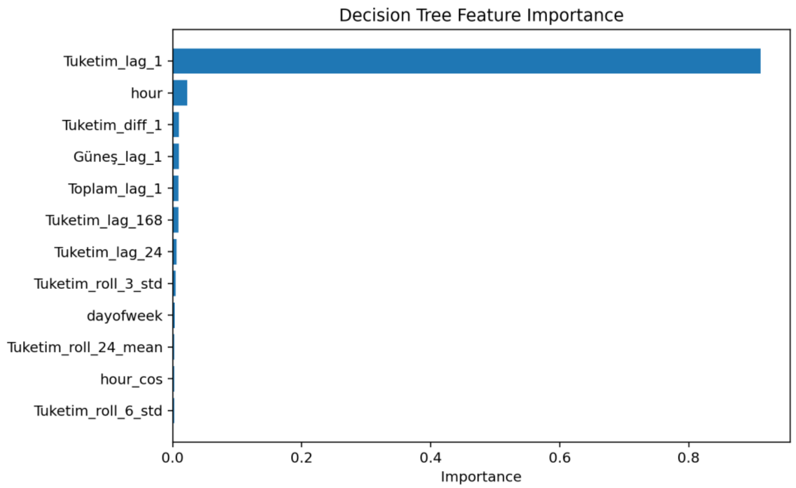
Split	Purpose
Earlier 85%	train + validation region
Last 15%	final test set
TimeSeriesSplit	hyperparameter tuning
Metrics	RMSE, MAE, R2

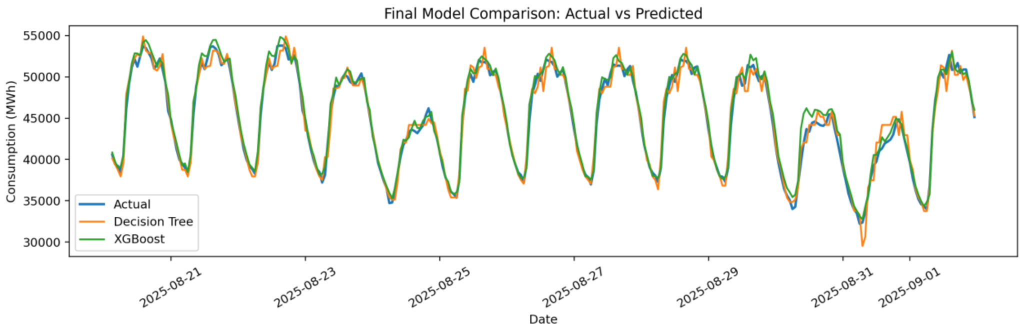
Random split is avoided because it can train on future samples and validate on past samples. TimeSeriesSplit preserves chronological order.

Model 1: Decision Tree Regressor

Interpretable baseline with controlled complexity

Baseline tree default settings → **Grid search max_depth, leaf size** → **TimeSeriesSplit 5 validation folds** → **Final test RMSE, MAE, R2**

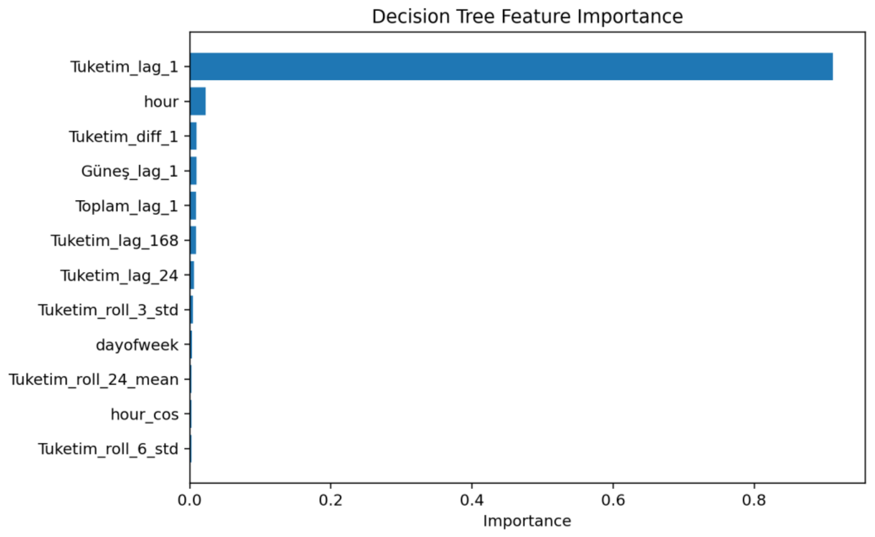




Decision Tree Results

Tuning reduces complexity but only slightly improves RMSE

Model	RMSE	MAE	R2
Baseline	1011.92	739.69	0.9688
Tuned	1004.10	759.15	0.9693



Best tuned configuration: max_depth=10, min_samples_leaf=5, min_samples_split=2.

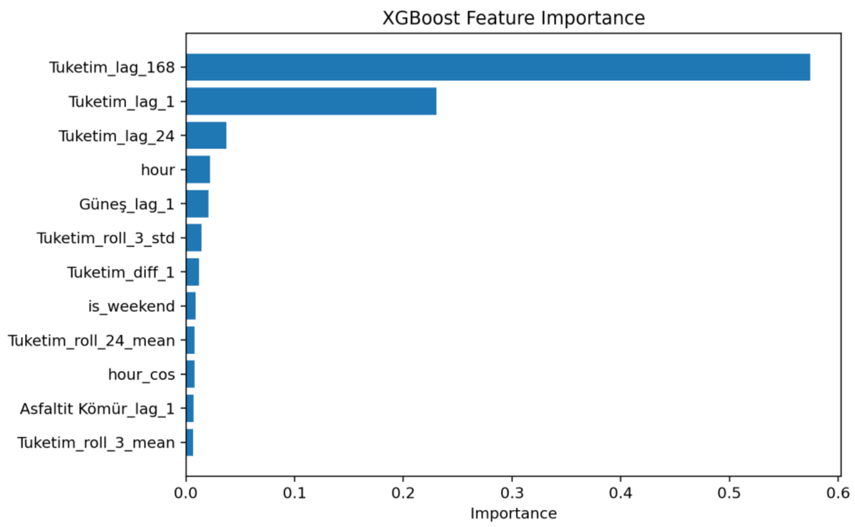
The model is interpretable, but it is still a single tree; boosted trees are expected to generalize better.

Model 2: XGBoost Regressor

Boosted tree ensemble for stronger nonlinear prediction

Sequential trees correct previous errors → **Small candidate search learning rate, depth, trees** → **Same TimeSeriesSplit no random CV** → **Same test set fair comparison**

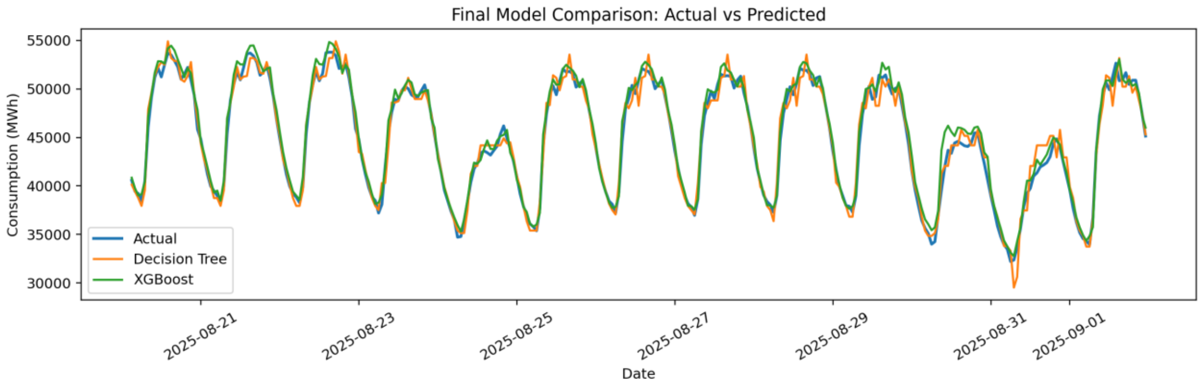
Final tuned parameters: 200 trees, depth 3, learning rate 0.03, subsample 0.9, colsample 0.9.



XGBoost Results

XGBoost achieves the lowest final test error

Model	RMSE	MAE	R2
Baseline	852.22	647.50	0.9779
Tuned	842.33	642.23	0.9784

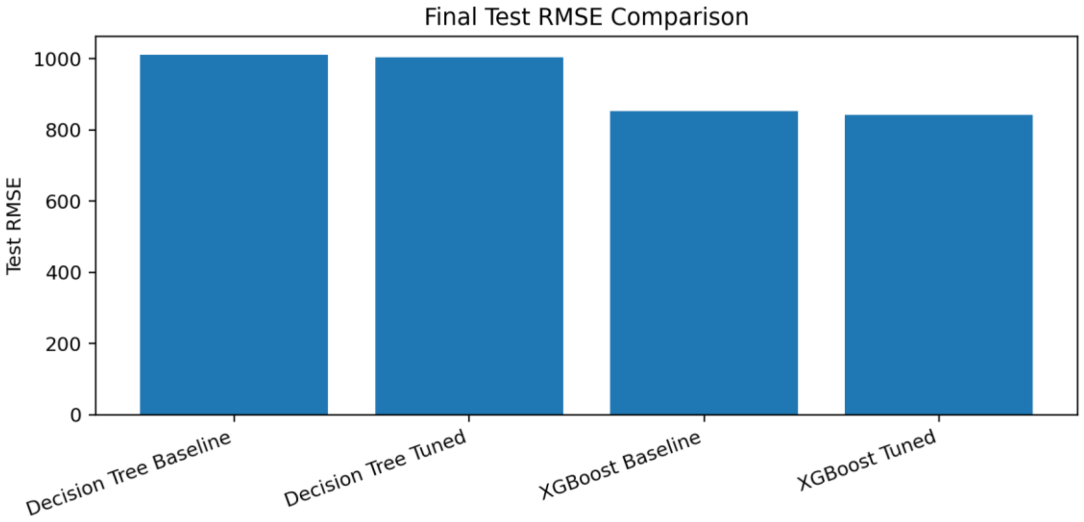


The tuned XGBoost model improves over the tuned Decision Tree by about 16.1% RMSE.

The remaining errors likely correspond to rapid local changes not captured by weather or holiday variables.

Final Model Comparison

Both models perform well; XGBoost is strongest



Model	RMSE	MAE	R2
Decision Tree Tuned	1004.10	759.15	0.9693
XGBoost Tuned	842.33	642.23	0.9784

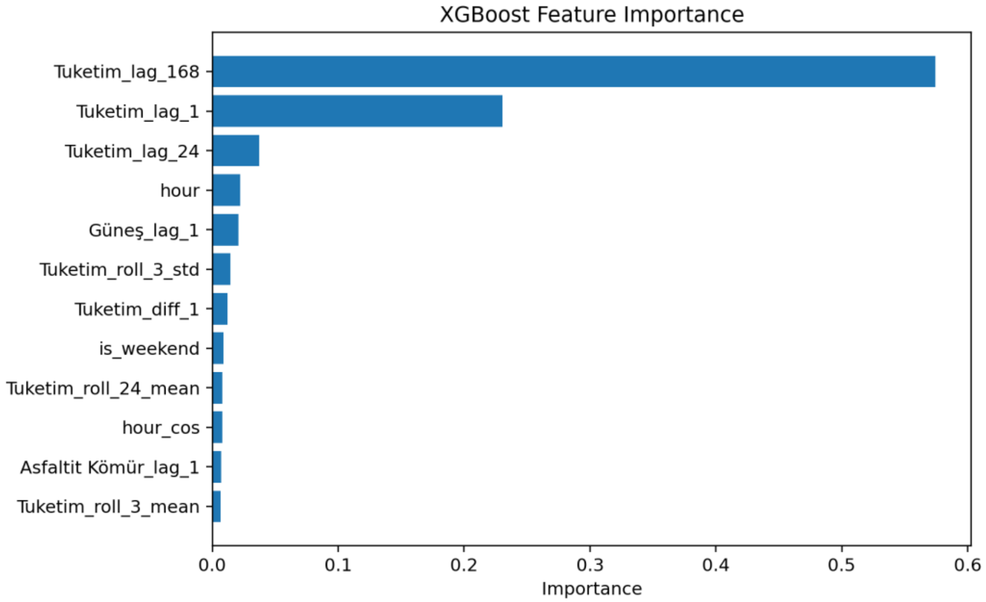
16.1%
RMSE reduction

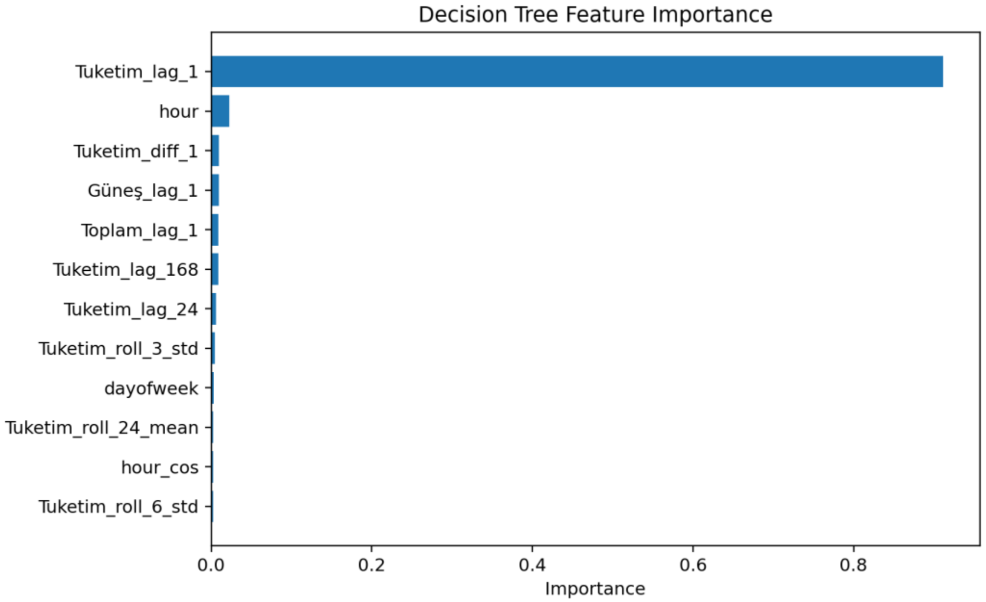
0.9784
Best R2

Final conclusion: XGBoost is selected as the better model under the same leakage-safe feature set and test period.

Insights and Interpretation

The strongest signals are recent consumption and system state





Lagged consumption features confirm strong short-term and daily dependence in electricity demand.

Lagged production features provide recent system-state information, but are kept lagged to avoid availability leakage.